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EX NO. 1

SETTING UP HYPERLEDGER FABRIC DEVELOPMENT ENVIRONMENT

AIM

To install and configure the Hyperledger Fabric development environment on a Linux system using Docker, Go, and Fabric binaries, create a blockchain test network, and establish a communication channel between peer organizations.

SOFTWARE REQUIREMENTS

- *Kali Linux / Ubuntu*
- *Docker*
- *Docker Compose*
- *Go (Version 1.20 or later)*
- *Git*
- *Curl*
- *Hyperledger Fabric Samples*
- *Hyperledger Fabric Docker Images*

HARDWARE REQUIREMENTS

- *Processor: Intel Core i3 or above*
- *RAM: Minimum 8 GB*
- *Storage: Minimum 20 GB free disk space*
- *Internet*

Connection PROCEDURE

Step 1: Update the System

Update the package repositories and upgrade the installed packages.

sudo apt update

sudo apt upgrade -y

Step 2: Install Required Packages

Install Git, Curl, jq, and Docker

Compose. sudo apt install -y git curl jq

docker-compose Verify installation:

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git --version curl --version

docker-compose --version

```
vboxuser@Ubuntu:~$ git --version
git version 2.43.0
vboxuser@Ubuntu:~$ curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.0 (+libidn2/2.3.7) libssh/0.10.6/openssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.12
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
```

Step 3: Verify

Docker Installation

Check the

installed Docker

version. docker --

version Start and enable Docker:

sudo systemctl start docker sudo

systemctl enable docker

Add the current user to the Docker group: sudo

usermod -aG docker \$USER newgrp docker

Verify Docker: docker run helloworld

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description: Ubuntu 24.04.4 LTS
Release: 24.04
Codename: noble
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3-24.04.2
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ apt-cache policy docker-ce docker-compose-plugin docker-compos
e
docker-ce:
  Installed: (none)
  Candidate: 5:29.7.2-1-ubuntu.24.04-noble
  Version table:
   5:29.7.2-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.7.1-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.7.0-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.6.2-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.6.1-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.6.0-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.5.3-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.5.2-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
   5:29.5.1-1-ubuntu.24.04-noble 500
      500 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages
```

```
vboxuser@Ubuntu:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3-24.04.2
vboxuser@Ubuntu:~$ sudo systemctl start docker
vboxuser@Ubuntu:~$ sudo systemctl enable docker
vboxuser@Ubuntu:~$ sudo usermod -aG docker $USER
vboxuser@Ubuntu:~$ newgrp docker
vboxuser@Ubuntu:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cddb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
```

Step 4: Verify Go

Installation Check

the installed Go

version. go version

Example Output:

go version go1.22.0 linux/amd64 Step 5:

Create a Workspace

Create a directory for the Hyperledger Fabric

project. mkdir -p ~/fabric-dev cd ~/fabric-dev

Verify the directory:

pwd

Step 6: Download Hyperledger Fabric Samples

Download the Hyperledger Fabric samples, binaries, and

Docker images. curl -sSL https://bit.ly/2ysbOFE | bash -s Wait until

all required files are downloaded successfully. Step 7: Navigate to

the Fabric Samples

Directory cd fabric-samples

```

vboxuser@Ubuntu:~$ go version
Command 'go' not found, but can be installed with:
sudo snap install go          # version 1.26.7, or
sudo apt install golang-go    # version 2:1.21-2
sudo apt install gccgo-go     # version 2:1.21-2
See 'snap info go' for additional versions.
vboxuser@Ubuntu:~$ mkdir -p ~/fabric-dev
vboxuser@Ubuntu:~$ cd ~/fabric-dev/
vboxuser@Ubuntu:~/fabric-dev$ pwd
/home/vboxuser/fabric-dev
vboxuser@Ubuntu:~/fabric-dev$ curl -sSL https://bit.ly/2ysb0FE | bash -s

Clone hyperledger/fabric-samples repo

====> Cloning hyperledger/fabric-samples repo
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15868, done.
remote: Counting objects: 100% (3446/3446), done.
remote: Compressing objects: 100% (631/631), done.
remote: Total 15868 (delta 3021), reused 2815 (delta 2815), pack-reused 12422 (from 2)
Receiving objects: 100% (15868/15868), 24.09 MiB | 4.99 MiB/s, done.
Resolving deltas: 100% (9146/9146), done.
fabric-samples v2.5.16 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

====> Downloading version 2.5.16 platform specific fabric binaries
====> Downloading: https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyperledger-fabric-linux-amd64-2.5.16.tar.gz

```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
0	0	0	0	0	0	0	0

Step 8: Configure Environment Variables Add the Fabric binaries to the system

```

PATH. echo 'export PATH=$PATH:$HOME/fabric-
dev/fabric-samples/bin' >>
~/.zshrc source ~/.zshrc

```

Step 9: Verify Hyperledger Fabric Installation

Verify the Peer binary:

peer version

Verify the Orderer binary: orderer version

```

hyperledger/fabric-orderer:2.5.16 e322c57331d3 182MB 50.7MB
hyperledger/fabric-orderer:latest e322c57331d3 182MB 50.7MB
hyperledger/fabric-peer:2.5 09ee75042de9 232MB 66.8MB
hyperledger/fabric-peer:2.5.16 09ee75042de9 232MB 66.8MB
hyperledger/fabric-peer:latest 09ee75042de9 232MB 66.8MB
vboxuser@Ubuntu:~/fabric-dev$ cd fabric-samples
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$ echo 'export PATH=$PATH:$HOME/fabric-dev/fabric-samples/bin' >> ~/.zshrc
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$ source ~/.zshrc
bash: /home/vboxuser/.zshrc: No such file or directory
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$ source ~/.zshrc
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$ peer version

Commit: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64
Chaincode:
Base Docker Label: org.hyperledger.fabric
Docker Namespace: hyperledger

vboxuser@Ubuntu:~/fabric-dev/fabric-samples$
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$ orderer version
orderer:
Version: v2.5.16
Commit SHA: f871cf9
Go version: go1.26.4
OS/Arch: linux/amd64

vboxuser@Ubuntu:~/fabric-dev/fabric-samples$

```

Step 10: Start the Hyperledger Fabric Test Network

Navigate to the test network directory. cd

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~/fabric-dev/fabric-samples/test-network

```
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$  
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$  
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$ orderer version  
orderer:  
Version: v2.5.16  
Commit SHA: f871cf9  
Go version: go1.26.4  
OS/Arch: linux/amd64  
  
vboxuser@Ubuntu:~/fabric-dev/fabric-samples$ cd ~/fabric-dev/fabric-samples/test-network  
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up  
Using docker and docker-compose  
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from  
'cryptogen'  
LOCAL_VERSION=v2.5.16  
DOCKER_IMAGE_VERSION=v2.5.16  
/home/vboxuser/fabric-dev/fabric-samples/test-network/./bin/cryptogen  
Generating certificates using cryptogen tool  
Creating Org1 Identities  
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations  
org1.example.com  
+ res=0  
Creating Org2 Identities  
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations  
org2.example.com  
+ res=0  
Creating Orderer Org Identities  
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations  
+ res=0  
Generating CCP files for Org1 and Org2  
Traceback (most recent call last):  
File "/usr/bin/docker-compose", line 22, in <module>
```

Start the Fabric test network:

`./network.sh up`

The command performs the following tasks:

- Generates cryptographic certificates.
- Creates the required Docker containers.
- Starts the Orderer.
- Starts the peer organizations.
- Creates the Fabric network.

```

from distutils.spawn import find_executable
ModuleNotFoundError: No module named 'distutils'
CONTAINER ID   IMAGE      COMMAND                  CREATED          STATUS          PORTS          NAMES
eed8d8ce6bc1   hello-world  "/hello"                9 minutes ago   Exited (0) 9 minutes ago           keen_haibt
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ which docker
which docker-compose
which docker-compose || true
docker compose version
/usr/bin/docker
/usr/bin/docker-compose
/usr/bin/docker-compose
Docker Compose version v5.5.0
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ls -l /usr/bin/docker-compose
-rwxr-xr-x 1 root root 996 Mar  9 2024 /usr/bin/docker-compose
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ sudo apt remove docker-compose
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  bridge-utils containerd docker.io libfwupd2 libgl1-amber-dri libglapi-mesa libwoff1 pigz python3-compose
  python3-docker python3-dockerpty python3-dockerpty python3-dotenv python3-packaging python3-texttable python3-websocket
  runc ubuntu-fan
Use 'sudo apt autoremove' to remove them.
The following packages will be REMOVED:
  docker-compose
0 upgraded, 0 newly installed, 1 to remove and 1 not upgraded.
After this operation, 61.4 kB disk space will be freed.
Do you want to continue? [Y/n] Y
(Reading database ... 198592 files and directories currently installed.)
Removing docker-compose (1.29.2-6ubuntu1) ...
Processing triggers for man-db (2.12.0-4build2) ...

```

```

Version table:
1.29.2-6ubuntu1 500
500 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 Packages
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up
Using docker and docker compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb'
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] up 7/7
✓ Network fabric_test Created 0.1s
✓ Volume compose_peer0.org2.example.com Created 0.0s
✓ Volume compose_orderer.example.com Created 0.0s
✓ Volume compose_peer0.org1.example.com Created 0.0s
✓ Container peer0.org2.example.com Started 2.9s
✓ Container peer0.org1.example.com Started 2.2s
✓ Container orderer.example.com Started 2.6s
CONTAINER ID   IMAGE      COMMAND                  CREATED          STATUS          PORTS          NAMES
73729b8a4930   hyperledger/fabric-peer:latest  "peer node start"    3 seconds ago   Up 2 seconds   0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp   peer0.org2.example.com
3ad79b31826e   hyperledger/fabric-orderer:latest  "orderer"            3 seconds ago   Up 2 seconds   0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp   orderer.example.com
ce42d655af3a   hyperledger/fabric-peer:latest  "peer node start"    3 seconds ago   Up 2 seconds   0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp   peer0.org1.example.com
eed8d8ce6bc1   hello-world  "/hello"                12 minutes ago   Exited (0) 12 minutes ago           keen_haibt

```

Step 11: Verify Running Containers

Display all active Docker

containers: `docker ps`

Expected containers include:

- `orderer.example.com`
- `peer0.org1.example.com`

- *peer0.org2.example.com*

```
ce42d655af3a hyperledger/fabric-peer:latest "peer node start" 3 seconds ago Up 2 seconds 0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
peer0.org1.example.com
eed8d8ce6bc1 hello-world "/hello" 12 minutes ago Exited (0) 12 minutes ago
keen_haibt
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
NAMES
73729b8a4930 hyperledger/fabric-peer:latest "peer node start" About a minute ago Up About a minute 0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tcp
peer0.org2.example.com
3ad79b31826e hyperledger/fabric-orderer:latest "orderer" About a minute ago Up About a minute 0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
orderer.example.com
ce42d655af3a hyperledger/fabric-peer:latest "peer node start" About a minute ago Up About a minute 0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
peer0.org1.example.com
vboxuser@Ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh createChannel -c mychannel
Using docker and docker compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb'
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] up 3/3
✓ Container peer0.org1.example.com Running 0.0s
✓ Container peer0.org2.example.com Running 0.0s
✓ Container orderer.example.com Running 0.0s
```

Step 12: Create a Channel

Create an application channel named mychannel.

./network.sh createChannel -c mychannel

This command:

- Generates the required channel configuration.
- Creates the channel.
- Joins both peer organizations to the channel.
- Configures the anchor peers.

Step 13: Verify Successful Channel Creation Successful messages

include:

Channel 'mychannel' created successfully

Successfully submitted proposal to join channel

Anchor peer set for Org1

Anchor peer set for Org2

Step 14: Stop the Network

After completing the experiment, stop the Fabric network.

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./network.sh down

FLOW OF THE SETUP

Install Dependencies

|



Install Docker & Go

|



Download Hyperledger Fabric

|



Configure Environment Variables

|



Verify Installation

|



Start Test Network

|



Create Channel

|



Join Peer Organizations

|



Configure Anchor Peers

|



Hyperledger Fabric Environment Ready

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RESULT

The Hyperledger Fabric development environment was successfully installed and configured. The required Fabric binaries and Docker images were installed, the test network was created successfully, the mychannel application channel was generated, both peer organizations joined the channel, and the anchor peers were configured successfully. The development environment is now ready for deploying chaincode (smart contracts) and developing blockchain applications.